# COMP 421 - Database Systems

Project 3: Application

Yazdan Zinati
yazdan.zinati@mail.mcgill.ca
260928077

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# 1 Relational Model

The relational model has not been modified from Project 2.

## Model

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`HealthcareInstitution(hcId, phoneNum, name, email, address, website)`

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`CommunityClinic(hcId)`

`hcId` foreign key referencing `HealthcareInstitution`

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`BirthingCenter(hcId)`

`hcId` foreign key referencing `HealthcareInstitution`

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`Midwife(pracId, name, email, phoneNum, hcId)`

`hcId` foreign key referencing `HealthcareInstitution`

note: `hcId` has to be `NOT NULL` to enforce Midwife's participation constraint in the `employs` relationship set.

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`Father(fatherId, phoneNum, name, email, address, QChcardid, dateOfBirth, profession, bloodType)`

---

`Mother(QChcardid, phoneNum, name, email, adddress, dateOfBirth, profession, bloodType)`

---

`Couple(coupleId, invited, interested, selected, QChcardid, fatherId)`

`QChcardid` foreign key referencing `Mother`
`fatherId` foreign key referencing `Father`

note: `QChcardid` has to be `NOT NULL` to enforce Couple's participation constraint in the `constitutesF` relationship set.

---

`InformationSession(sessId, language, date, time, pracId)`

`pracId` foreign key referencing Midwife

note: `pracId` has to be `NOT NULL` to enforce InformationSession's participation constraint in the `hosts` relationship set.

---

`Invitation(sessId, coupleId, attended, registered)`

`sessId` foreign key referencing `Information Session`
`coupleId` foreign key referencing `Couple`

---

`Pregnancy(coupleId, pregNum, estDueDate, expBirthTimeframe, lastMenstPeriod, isHomeBirth, primaryMidwife, backupMidwife)`

`coupleId` foreign key referencing `Couple`
`primaryMidwife` foreign key referencing `Midwife(pracId)`
`backupMidwife` foreign key referencing `Midwife(pracId)`

note: `coupleId` is part of the primary key and thus has to be `NOT NULL`.

---

Appointment(apptId, date, time, pregNum, coupleId, pracId)

(pregNum, coupleId) foreign key referencing Pregnancy
pracId foreign key referencing Midwife

note: pregNum and coupleId has to be NOT NULL to enforce Appointment's participation constraint in the belongs relationship set.

note: pracId has to be NOT NULL to enforce Appointment's participation constraint in the sets up relationship set.

---

Note(noteId, timestamp, description, apptId)

apptId foreign key referencing Appointment

note: apptId has to be NOT NULL to enforce Note's participation constraint in the belongs relationship set.

---

Baby(babyId, name, gender, timeOfBirth, dateOfBirth, bloodType, pregNum, coupleId)

(pregNum, coupleId) foreign key referencing Pregnancy

note: pregNum and coupleId has to be NOT NULL to enforce Baby's participation constraint in the born from relationship set.

---

Technician(techId, name, phoneNum)

---

Test(testId, type, datePrescribed, dateSampleTaken, results, dateConducted, lab, techId, pracId, pregNum, coupleId, babyId, QChcardid)

techId foreign key referencing Technician
pracId foreign key referencing Midwife
(pregNum, coupleId) foreign key referencing Pregnancy
babyId foreign key referencing Baby
QChcardid foreign key referencing Mother

note: pregNum, coupleId has to be NOT NULL to enforce Test's participation constraint in the belongs relationship set.

note: techId has to be NOT NULL to enforce Test's participation constraint in the conducts relationship set.

note: pracId has to be NOT NULL to enforce Test's participation constraint in the prescribes relationship set.

---

BirthLocation(hcId, pregNum, coupleId) hcId foreign key referencing BirthingCenter

(pregNum, coupleId) foreign key referencing Pregnancy

---

**Note: relation names have been pluralized in the db tables to follow convention.**

| Entity Set | Artificial key |
|---|---|
| HealthcareInstitution | hcId |
| Father | fatherId |
| InformationSession | sessId |
| Test | testId |
| Baby | babyId |
| Appointment | apptId |
| Note | noteId |
| Couple | coupleId |

Table 1: Artificial keys created in the relational model

### Artificial keys

## 2 Application Interaction

a. **Assume: `P0` = 123.**

i Run a query in the database that shows that midwife with practitioner id `P0` does not exist.

```
db2 => SELECT name FROM Midwives WHERE pracId = 123;

NAME
------------------------------

  0 record(s) selected.
```

Figure 1: a (i)

ii Start the application and enter the practitioner id (of the midwife) `P0`. **ii.**

```
szinat@winter2022-comp421:~$ java GoBabbyApp
----- Go Babby Application ------

Please enter your practitioner id [E] to exit: 123
Invalid practitioner id

Please enter your practitioner id [E] to exit:
```

Figure 2: a (ii)

b. **Assume:**

- **mother `M1` is named *Victoria Gutierrez***
- **mother `M2` is named *Corina Solis***
- **Appointment date `D1` = 2022/03/21.**
- **Note `N1`'s observation is *This is my newest note.***

i Run a query in the database that shows that a mother M1 has had at least two pregnancies (can be as part of the same or different couple - does not matter).

```
db2 => SELECT C.COUPLEID, PREGNUM, NAME, C.QCHCARDID, PRIMARYMIDWIFE, BACKUPMIDWIFE FROM PREGNANCIES
JOIN COUPLES C on PREGNANCIES.COUPLEID = C.COUPLEID
JOIN MOTHERS M on C.QCHCARDID = M.QCHCARDID
WHERE NAME='Victoria Gutierrez';
db2 (cont.) => db2 (cont.) => db2 (cont.) =>
COUPLEID    PREGNUM     NAME                           QCHCARDID    PRIMARYMIDWIFE BACKUPMIDWIFE
----------- ----------- ------------------------------ ------------ -------------- -------------
       4010           1 Victoria Gutierrez             VIGU63128923           1005          1002
       4010           2 Victoria Gutierrez             VIGU63128923           1010          1001
       4010           3 Victoria Gutierrez             VIGU63128923           1001          1002

  3 record(s) selected.
```

Figure 3: b (i)

ii In the application, enter the practitioner id (of the midwife) `P1` (must exist in the database). Assume `P1` = 1001.

```
szinat@winter2022-comp421:~$ java GoBabbyApp
----- Go Babby Application ------

Please enter your practitioner id [E] to exit: 1001
Welcome Marion

Please enter the date (YYYY-MM-DD) for appointment list [E] to exit:
```

Figure 4: b (ii)

iii Enter the date `D1`. One of the appointments must be for `M1` and another one for `M2` , a different expecting mother.

```
szinat@winter2022-comp421:~$ java GoBabbyApp
----- Go Babby Application ------

Please enter your practitioner id [E] to exit: 1001
Welcome Marion

Please enter the date (YYYY-MM-DD) for appointment list [E] to exit: 2022-03-21
1:      07:45:00 B Victoria Gutierrez VIGU63128923
2:      10:35:00 B Corina Solis CASL23451234
3:      18:10:00 P Judith Leon JULE09156234

Enter the appointment number that you would like to work on.
        [E] to exit [D] to go back to another date:
```

Figure 5: b (iii)

iv Choose the appointment number associated with `M1`. TODO

v Enter 1. (Should show at least two notes).

```
Enter the appointment number that you would like to work on.
        [E] to exit [D] to go back to another date: 1

For Victoria Gutierrez VIGU63128923

1. Review notes
2. Review tests
3. Add a note
4. Prescribe a test
5. Go back to the appointments

Enter your choice: 1
2022-02-13 12:23:32.0   Couple would prefer home birth.
2022-02-13 10:02:45.26  Baby has good movements

For Victoria Gutierrez VIGU63128923

1. Review notes
2. Review tests
3. Add a note
4. Prescribe a test
5. Go back to the appointments
```

Figure 6: b (v)

vi Enter 2. (Should show at least two tests).

```
For Victoria Gutierrez VIGU63128923

1. Review notes
2. Review tests
3. Add a note
4. Prescribe a test
5. Go back to the appointments

Enter your choice: 2
2020-01-23 [blood iron] 75 mcg/dL
2020-01-23 [blood iron] 86 mcg/dL

For Victoria Gutierrez VIGU63128923

1. Review notes
2. Review tests
3. Add a note
4. Prescribe a test
5. Go back to the appointments

Enter your choice:
```

Figure 7: b (vi)

vii Enter 3. Add a note `N1`.

```
For Victoria Gutierrez VIGU63128923

1. Review notes
2. Review tests
3. Add a note
4. Prescribe a test
5. Go back to the appointments

Enter your choice: 3
Please type your observation: This is my newest note.

For Victoria Gutierrez VIGU63128923

1. Review notes
2. Review tests
3. Add a note
4. Prescribe a test
5. Go back to the appointments

Enter your choice:
```

Figure 8: b (vii)

viii Run a query in the database that shows all the notes associated with M1 , include couple information, pregnancy number, etc., as relevant in your model. The actual note text may be truncated to first 50 characters or so to reduce clutter.

```
db2 => SELECT A.COUPLEID, A.PREGNUM, NAME, C.QCHCARDID, PRACID, CAST (DESCRIPTION AS VARCHAR(35))
FROM NOTES
        JOIN APPOINTMENTS A on NOTES.APPTID = A.APPTID
        JOIN PREGNANCIES P on A.PREGNUM = P.PREGNUM and A.COUPLEID = P.COUPLEID
        JOIN COUPLES C on P.COUPLEID = C.COUPLEID
        JOIN MOTHERS M on C.QCHCARDID = M.QCHCARDID
WHERE NAME = 'Victoria Gutierrez';db2 (cont.) => db2 (cont.) => db2 (cont.) => db2 (cont.) => db2 (cont.) => db2 (cont.) =>

COUPLEID    PREGNUM     NAME                           QCHCARDID    PRACID      6
----------- ----------- ------------------------------ ------------ ----------- ----------------------------------
       4010           2 Victoria Gutierrez             VIGU63128923        1001 Couple would prefer home birth.
       4010           2 Victoria Gutierrez             VIGU63128923        1001 This is my newest note.
       4010           2 Victoria Gutierrez             VIGU63128923        1001 Baby has good movements

  3 record(s) selected.
```

Figure 9: b (viii)

ix Enter 1.

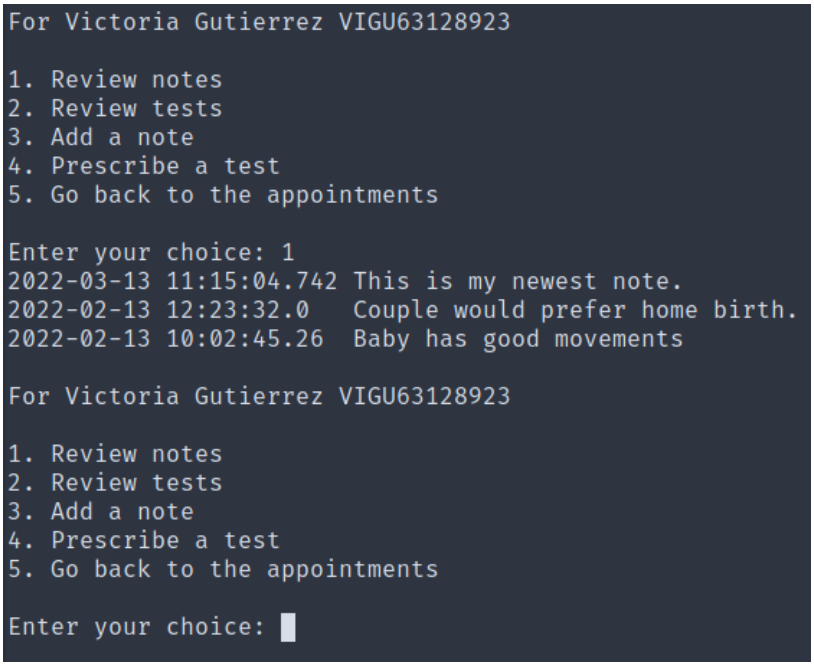



Figure 10: b (viii)

x Enter 4. Add a test for trisomy 21.

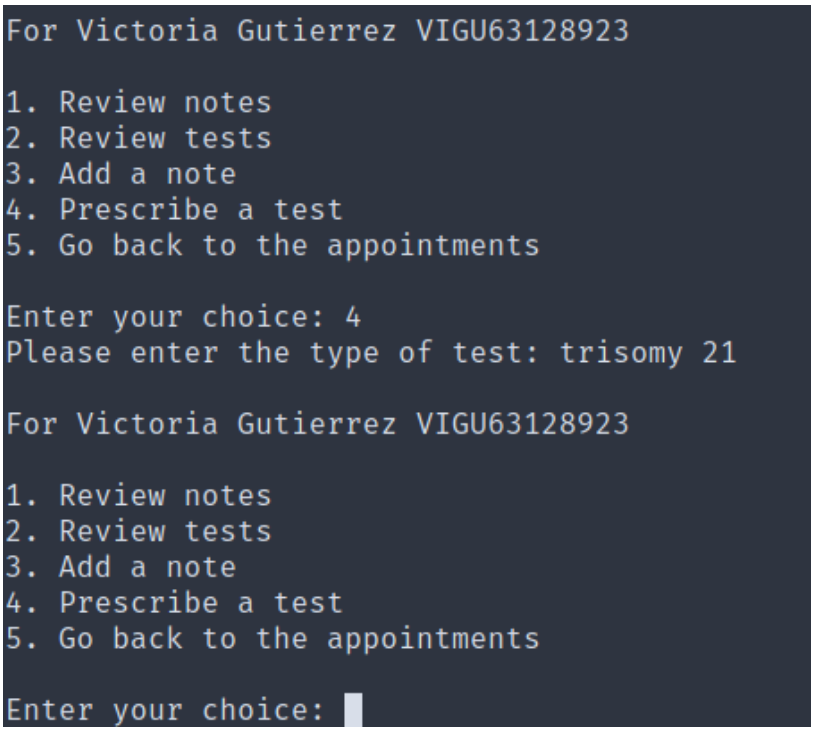



Figure 11: b (x)

xi Run a query in the database that shows all the tests associated with `M1` , include couple information, pregnancy number, etc., as relevant in your model. The result attribute of the test maybe truncated to the first 50 characters or so.

```
db2 => SELECT TYPE, NAME, M.QCHCARDID, COUPLEID, PREGNUM, CAST(RESULTS AS VARCHAR(35))
FROM TESTS
        JOIN MOTHERS M on TESTS.QCHCARDID = M.QCHCARDID
WHERE NAME = 'Victoria Gutierrez';db2 (cont.) => db2 (cont.) => db2 (cont.) =>

TYPE                          NAME                          QCHCARDID    COUPLEID    PREGNUM     6
----------------------------- ----------------------------- ------------ ----------- ----------- ------------
trisomy 21                    Victoria Gutierrez            VIGU63128923        4010           2 -
blood iron                    Victoria Gutierrez            VIGU63128923        4010           2 75 mcg/dL
blood iron                    Victoria Gutierrez            VIGU63128923        4010           2 86 mcg/dL

  3 record(s) selected.
```

Figure 12: b (xi)

xii Enter 5 TODO

c. **Assume: mother `M2` is named *Corina Solis***

i Run a query in the database that shows all the tests associated with `M2` , include couple information, pregnancy number, etc., as relevant in your model. The result attribute of the test maybe truncated to the first 50 characters or so.

```
db2 => SELECT TYPE, NAME, M.QCHCARDID, COUPLEID, PREGNUM, CAST(RESULTS AS VARCHAR(35)) AS RESULTS
FROM TESTS
        JOIN MOTHERS M on TESTS.QCHCARDID = M.QCHCARDID
WHERE NAME = 'Corina Solis';
db2 (cont.) => db2 (cont.) => db2 (cont.) =>
TYPE                          NAME                          QCHCARDID    COUPLEID    PREGNUM     RESULTS
----------------------------- ----------------------------- ------------ ----------- ----------- ------------
blood iron test               Corina Solis                  CASL23451234        4002           1 -
sonography                    Corina Solis                  CASL23451234        4002           1 -

  2 record(s) selected.
```

Figure 13: c (i)

ii Choose the appointment associated with mother `M2`

```
For Victoria Gutierrez VIGU63128923

1. Review notes
2. Review tests
3. Add a note
4. Prescribe a test
5. Go back to the appointments

Enter your choice: 5
1:      07:45:00 B Victoria Gutierrez VIGU63128923
2:      10:35:00 B Corina Solis CASL23451234
3:      18:10:00 P Judith Leon JULE09156234

Enter the appointment number that you would like to work on.
       [E] to exit [D] to go back to another date: 2

For Corina Solis CASL23451234

1. Review notes
2. Review tests
3. Add a note
4. Prescribe a test
5. Go back to the appointments

Enter your choice:
```

Figure 14: c (ii)

iii Enter 1 (to see the relevant notes).

```
For Corina Solis CASL23451234

1. Review notes
2. Review tests
3. Add a note
4. Prescribe a test
5. Go back to the appointments

Enter your choice: 1
2022-03-13 09:24:05.26  Everything is going well yay!
2022-03-12 20:07:17.658 this is great!

For Corina Solis CASL23451234

1. Review notes
2. Review tests
3. Add a note
4. Prescribe a test
5. Go back to the appointments

Enter your choice:
```

Figure 15: c (iii)

iv Enter 4. Add a test prescription for Group B streptococcus (test type).

```
For Corina Solis CASL23451234

1. Review notes
2. Review tests
3. Add a note
4. Prescribe a test
5. Go back to the appointments

Enter your choice: 4
Please enter the type of test: Group B streptococcus

For Corina Solis CASL23451234

1. Review notes
2. Review tests
3. Add a note
4. Prescribe a test
5. Go back to the appointments

Enter your choice:
```

Figure 16: c (iv)

v Run a query in the database that shows all the tests associated with `M2` , include couple information, pregnancy number, etc., as relevant in your model. The result attribute of the test maybe truncated to the first 50 characters or so.

```
db2 => SELECT TYPE, NAME, M.QCHCARDID, COUPLEID, PREGNUM, CAST(RESULTS AS VARCHAR(35)) AS RESULTS
FROM TESTS
         JOIN MOTHERS M on TESTS.QCHCARDID = M.QCHCARDID
WHERE NAME = 'Corina Solis';
db2 (cont.) => db2 (cont.) => db2 (cont.) =>
TYPE                           NAME                           QCHCARDID    COUPLEID    PREGNUM     RESULTS
------------------------------ ------------------------------ ------------ ----------- ----------- --------
Group B streptococcus          Corina Solis                   CASL23451234        4002           1 -
blood iron test                Corina Solis                   CASL23451234        4002           1 -
sonography                     Corina Solis                   CASL23451234        4002           1 -

  3 record(s) selected.
```

Figure 17: c (v)

d. **Assume appointment date `D2` = 2022/05/29**

i Choose 5, followed by D.

```
For Corina Solis CASL23451234

1. Review notes
2. Review tests
3. Add a note
4. Prescribe a test
5. Go back to the appointments

Enter your choice: 5
1:      07:45:00 B Victoria Gutierrez VIGU63128923
2:      10:35:00 B Corina Solis CASL23451234
3:      18:10:00 P Judith Leon JULE09156234

Enter the appointment number that you would like to work on.
        [E] to exit [D] to go back to another date: d

Please enter the date (YYYY-MM-DD) for appointment list [E] to exit:
```

Figure 18: d (i)

ii Enter a different date D2

```
Enter your choice: 5
1:      07:45:00 B Victoria Gutierrez VIGU63128923
2:      10:35:00 B Corina Solis CASL23451234
3:      18:10:00 P Judith Leon JULE09156234

Enter the appointment number that you would like to work on.
        [E] to exit [D] to go back to another date: d

Please enter the date (YYYY-MM-DD) for appointment list [E] to exit: 2022-05-29
1:      07:45:00 B Judith Leon JULE09156234
2:      08:45:00 B Bobbi Cherry BOCH07562649

Enter the appointment number that you would like to work on.
        [E] to exit [D] to go back to another date:
```

Figure 19: d (ii) and d (iii)

iii It should show at least two appointments for mothers M3 , M4 , who are not the same as previous examples.

# 3 Indexing

Create an index that will help retrieve a mother's address, assuming that we only have a phone number to start the search with.

```
db2 => CREATE UNIQUE INDEX idx1 -- the phoneNum has to be unique for indexing to be meaningful here
    ON MOTHERS (PHONENUM) INCLUDE (ADDRESS);db2 (cont.) =>
DB20000I  The SQL command completed successfully.
db2 =>
```

Figure 20: Q4

# 4 Data Analytics

```
db2 => EXPORT TO "/home/2022/szinat/birth_months.csv" OF DEL
SELECT *
FROM (
        SELECT SUBSTRING('JAN FEB MAR APR MAY JUN JUL AUG SEP OCT NOV DEC ', monthNo * 4 - 3, 3) AS birthMonth,
               numBirth
        FROM (
                SELECT MONTH(DATEOFBIRTH) AS monthNo, COUNT(DATEOFBIRTH) AS numBirth
                FROM BABIES
                WHERE DATEOFBIRTH IS NOT NULL
                GROUP BY MONTH(DATEOFBIRTH)
                ORDER BY MONTH(DATEOFBIRTH)
            )
);db2 (cont.) => db2 (cont.) => db2 (cont.) => db2 (cont.) => db2 (cont.) => db2 (cont.) => db2 (cont.) => db2 (cont.) => db2
 (cont.) => db2 (cont.) => db2 (cont.) => db2 (cont.) =>
SQL3104N  The Export utility is beginning to export data to file
"/home/2022/szinat/birth_months.csv".

SQL3105N  The Export utility has finished exporting "12" rows.


Number of rows exported: 12
```

Figure 21: SQL export to export db query into csv file

Plot generated using matplotlib. submitted python code: `data_analytics.py`.

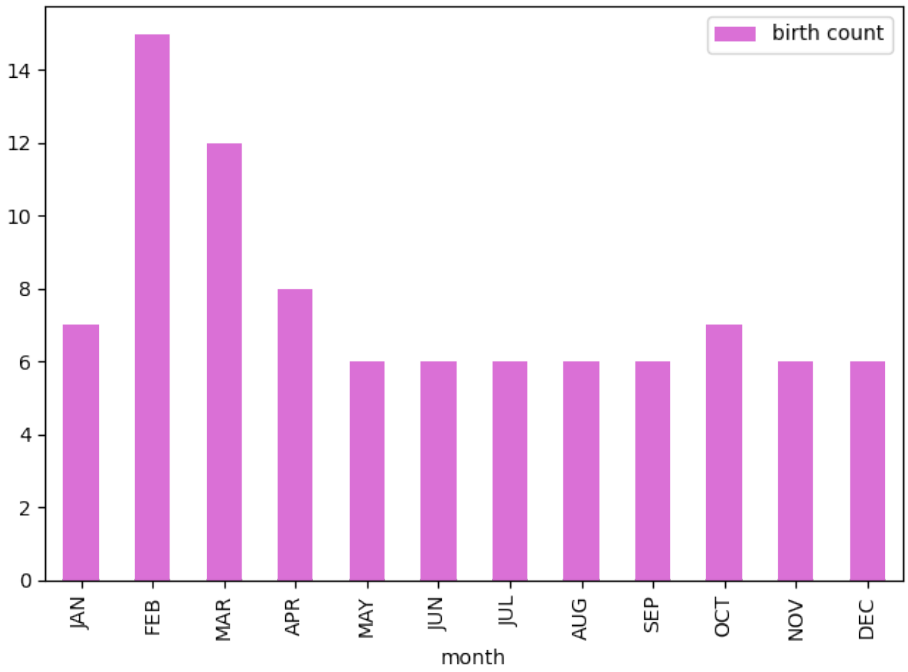



Figure 22: Bar plot showing the number of births in each month